

lab 2, mark (need submit), 10%

- Jan 29 11:59 pm)

- Submit through BrightSpace (No email)

- ONE PDF File ($\leq 5MB$)

logic

1.

2.

3.

Name

Hyp.

D. Morgan

Modus Pares

~~E1~~

~~E17~~

exmp: $A = \{1, 3, 5, 6\}$ $B = \{3, 4, 5, 6, 7\}$

$$R \subseteq A \times B$$

①

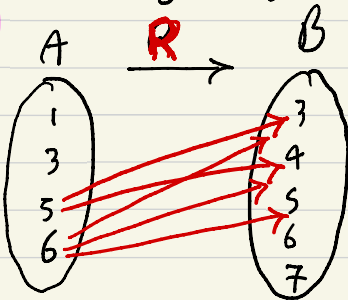
$$(x R y) \iff x > y$$

②

$$R = \{(5, 3), (5, 4), (6, 3), (6, 4), (6, 5)\}$$

③

arrowing diagram



④

relation matrix

$$R = \begin{matrix} & \begin{matrix} A & B \end{matrix} \begin{matrix} \downarrow \\ 1 \\ 3 \\ 5 \\ 6 \end{matrix} \end{matrix} \begin{matrix} \begin{matrix} 3 & 4 & 5 & 6 & 7 \end{matrix} \\ \left[\begin{array}{ccccc} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \end{array} \right] \end{matrix}$$

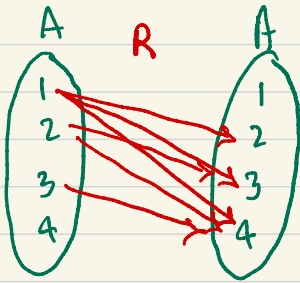
a relation on A : a subset of $A \times A$

$\uparrow$ domain $\uparrow$ codomain

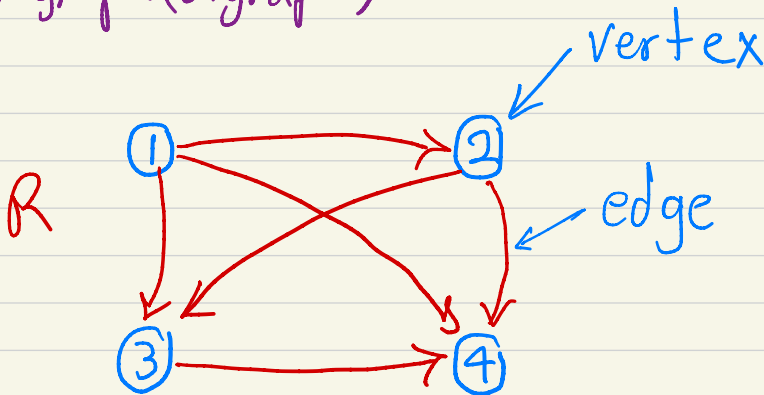
exmp: $A = \{1, 2, 3, 4\}$

$$R \subseteq A \times A$$

$$x R y \Leftrightarrow x < y$$



when domain = codomain we can represent R by a directed graph (digraph)



Properties of R on A :

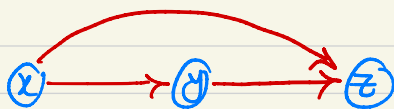
- R is reflexive if $\forall x \in A (x R x)$



- R is symmetric if $\forall x, y \in A (x R y \Rightarrow y R x)$



- R is transitive if $\forall x, y, z (x R y \wedge y R z \Rightarrow x R z)$



- R is antisymmetric if $\forall x, y \in A (x R y \Rightarrow y \not R x)$

$$\forall x, y \in A (x R y \wedge y R x \Rightarrow x = y)$$



exmp: $A = \{1, 2, 3, 4\}$

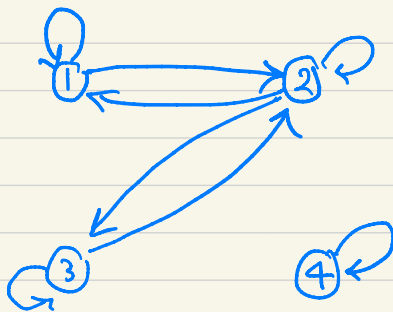
$R = \{(1, 1), (1, 2), (3, 2), (2, 2), (2, 3), (3, 3), (4, 4), (2, 1)\}$

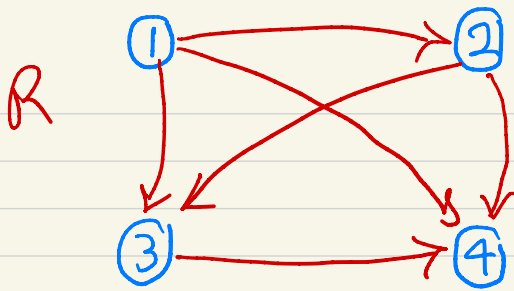
- Is R reflexive? Yes

- Is R symmetric? Yes

- Is R antisymmetric? No

- Is R transitive? No $(1, 2), (2, 3)$ but $(1, 3)$





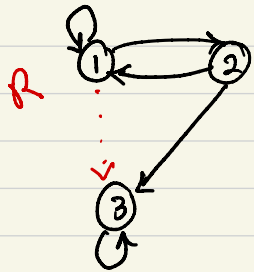
$$R = \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$$

reflexive? NO

symmetric? NO

antisymmetric? YES

transitive? YES



reflexive NO because $(2, 2) \notin R$

symmetric NO because $(2, 3) \in R$ but $(3, 2) \notin R$

antisymmetric NO because $(1, 2) \in R$ and $(2, 1) \in R$ but $1 \neq 2$
 $[(1, 2) \in R \text{ and } (2, 1) \in R]$

transitive NO because $(1, 2) \in R$ and $(2, 3) \in R$ but $(1, 3) \notin R$

think of a relation that is both symmetric and antisymmetric.

equivalence relation : is a relation that is reflexive, symmetric, and transitive.

exmp: $I_s \leq$ on $A = \{2, 3, 4, 5\}$ an equivalence relation?

$\hookrightarrow x$ is related to y iff $x \leq y$

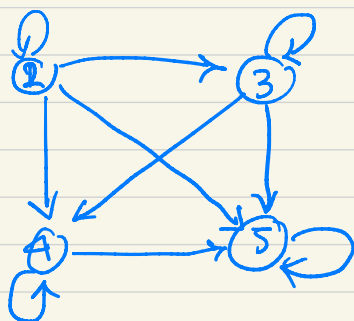
reflexive: $\forall x \in A (x \leq x)$ ✓

symmetric: No because $2 \leq 3$ but $3 \not\leq 2$

$\Rightarrow \leq$ is not symmetric so it is not an equivalence relation

transitive: $\forall x, y, z (x \leq y \wedge y \leq z \Rightarrow x \leq z)$ ✓

antisymmetric: $\forall x, y \in A (x \leq y \wedge y \leq x \Rightarrow x = y)$ ✓



Congruence: shows that two integers have the same remainder when divided by a given integer n .

$$a \equiv b \pmod{n} \quad \text{iff} \quad \text{remainder}(a, n) = \text{remainder}(b, n)$$

exmp:

$$a=17, \quad b=5, \quad n=3$$

$$\left. \begin{array}{l} 17 \bmod 3 = 2 \\ 5 \bmod 3 = 2 \end{array} \right\} \Rightarrow 17 \equiv 5 \pmod{3}$$

$$a=41, \quad b=17, \quad n=12$$

$$\left. \begin{array}{l} 41 \bmod 12 = 5 \\ 17 \bmod 12 = 5 \end{array} \right\} \Rightarrow 41 \equiv 17 \pmod{12}$$

if $a \equiv b \pmod{n}$ then $a = b + \underset{\substack{\uparrow \\ \text{some integer}}}{k} n$ and vice versa

$$17 \equiv 5 \pmod{3}$$

$$17 = 5 + k \times 3$$

$\uparrow$
4

$$41 \equiv 17 \pmod{12}$$

$$41 = 17 + k \times 12$$

$\uparrow$
2

exmp: $2 \equiv -3 \pmod{5}$

Remainder is positive

$$-3 \equiv -8 \pmod{5}$$

domain $\mathbb{Z}$
Is $\equiv$ an equivalence relation?